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Applications of Geospatial Analysis in different Areas and critical evaluation of “Geospatial analysis of wind energy plant in Saudi Arabia using a GIS-AHP technique”.

1.1: Critical Review of "Geospatial analysis of wind energy plant in Saudi Arabia using a GIS-AHP technique”.

1.1.1. Summary of Key Techniques:

The paper forms a study that integrates Geographic Information System (GIS) and Analytical Hierarchy Process (AHP), which is an MCDM method, to solve problems of wind farm siting in the Kingdom of Saudi Arabia (Albraheem and AlAwlaqi 2023).

- GIS: Manages, processes, and visualizes spatial data layers being criteria (wind speed, slope, and proximity) and constraints (airports and protected areas). Spatial operations included buffering, distance calculation, slope derivation, and reclassification (standardization of scores 1 to 4); lastly, the weighted overlay combined all layers according to their importance to produce the final suitability map (Albraheem and AlAwlaqi 2023, pp. 5882, 5884, 5887).

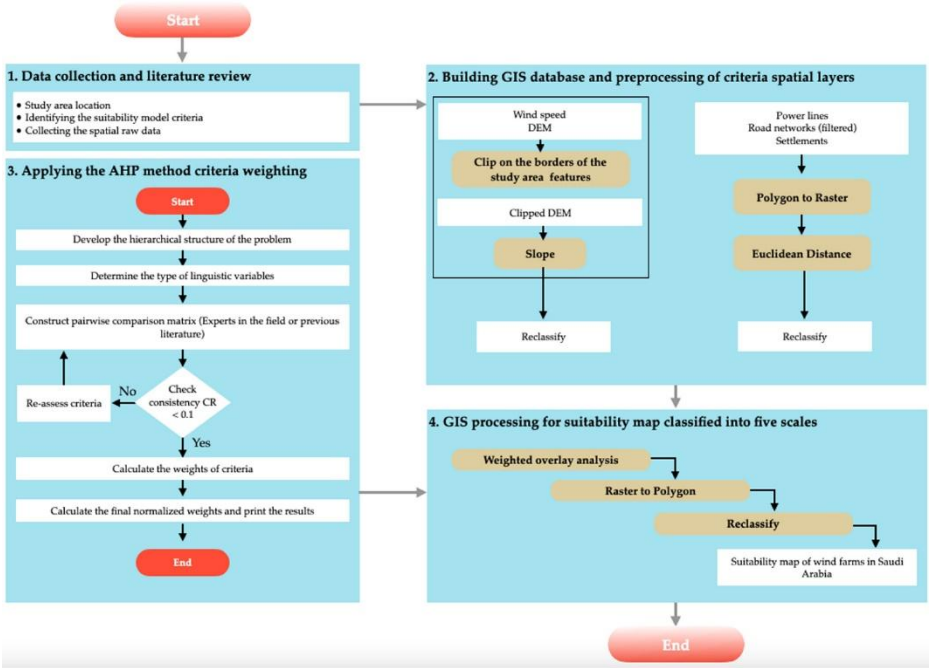

Figure 1: This illustrates the proposed framework (Albraheem and AlAwlaqi, 2023).

- **AHP:** decides the relative weights of evaluation criteria by hierarchically structuring the decision, using pairwise comparisons (Saaty's scale), deriving the weights from the principal eigenvector of a matrix, and checking for consistency of judgments (CR < 0.10). Wind speed was given the highest weight (53.9%), followed by grid proximity (23.7%), roads (9%), settlements (7.8%), and slope (5.7%) (Albraheem and AlAwlaqi, 2023, pp. 5886-5887).

1.1.2. Major Contributions:

1. **Enhanced Saudi Suitability Model:** Addresses the imperfections of previous works through the inclusion of criteria that earlier had been left out (slope, water bodies, IBAs), a wider range of wind speeds, and variable weights in AHP, considered to be a better way of assigning weights than the equal one (Albraheem and AlAwlaqi, 2023, pp. 5879, 5880, 5893).
2. **Detailed Site Identification:** Provides a national suitability map classifying land into four levels, quantifying areas for each level (e.g., 48,995 km² Extremely Suitable), and generating detailed maps at the regional level that present provincial-level potentials and coordinates of large-scale sites (Albraheem and AlAwlaqi, 2023, pp. 5889, 5890, 5893).

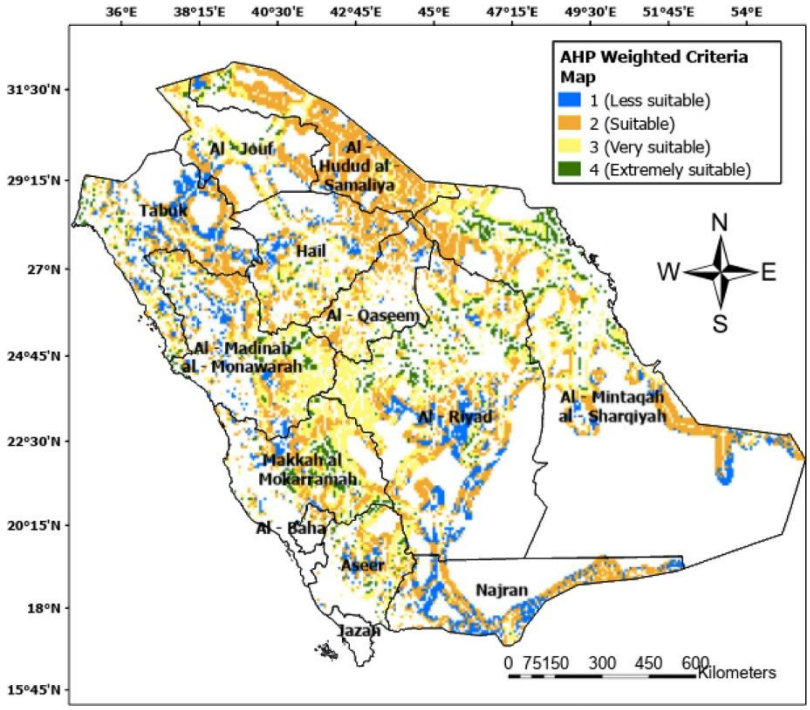

Figure 2: This shows the final wind suitability map for Saudi Arabia (Albraheem and AlAwlaqi, 2023).

3. **Practical Validation:** This validation of findings links lands the locations known for existing wind projects (Dumat Al Jandal) and announced wind projects in Saudi Arabia into support of the model as a decision-support tool (Albraheem and AlAwlaqi, 2023, p. 5893).

1.1.3. Critical Points:

1. **AHP Weight Provenance:** AHP weights come from the literature involving AHP for other countries (Albraheem and AlAwlaqi, 2023, p. 5886). This brings in doubt the potential fixation on non-transferable context-based weighing since factor importance can differ in Saudi Arabia. A sensitivity analysis on altering weight was not offered.
2. **Static Analysis:** It is a snapshot obtained from the available data (Albraheem & AlAwlaqi, 2023). It does not consider temporal changes either from fast-tracking infrastructure development or the land cover changes due to the mega Saudi Vision 2030 project, and maybe even long-term modifications in the wind regime from the global climate change.
3. **Uncertainty and data resolution:** Data sources such as Global Wind Atlas 292m resolution data and vector datasets at 1:50,000 from OpenStreetMap were used for the study (Albraheem and AlAwlaqi, 2023, p. 5884). They have neither explicitly discussed any possible uncertainties the data resolution may induce (which may conceal information about wind variability at the local scale) nor discussed about their accuracy or age.

1.2: Importance of Geospatial Analysis in Different Areas

Any location data get converted into information that is actionable by geospatial analysis, which is indispensable in many industries. The spatial context provides decision making and distribution of resources. Urban planning and environmental management are probably most impacted and majorly important.

1.2.1. Urban Planning:

- **Present Status:** Presently, with urban planning and smart cities, geospatial analytics analyze population, land use, zoning, transportation, and infrastructure (Esri, n.d.-a; IGNESA, 2024). Among the tools used in the process are mapping, spatial statistics, network analyses, and digital twins (Esri, n.d.-b; alicethompson, 2024).
- **Issues:** Heterogeneous, large-volume data integration; accuracy and timeliness assurance; dealing with Big Data; tool usability by different stakeholders; and privacy issues (Brown et al., 2013; arXiv, 2024a).
- **Solutions:** Advanced GIS software (e.g., ArcGIS), AI/ML for forecasting and pattern prediction, cloud computing for data management, 3D modelling (e.g., MetroVista™), and standardization efforts (Esri, n.d.-b; alicethompson, 2024; arXiv, 2024a).
- **Future Directions:** Changes in more reliance on real-time Internet of Things (IoT) data; advanced predictive GeoAI; richer AR/VR visualizations; and greater focus on social equity in planning (IGNESA, 2024; MGISS, 2024).

1.2.2. Environmental Management:

- **Current Status:** Crucial for resource monitoring, environmental impact assessment, and risk

management (Arun and Kumar, 2024; Tu et al., 2023). Uses include mapping of resources, biodiversity conservation, water resources management, climate change impact assessment, pollution monitoring, and disaster management, in which much dependence is put on remote sensing (Costa et al., 2024; Dritsas and Trigka, 2025).

- **Challenges:** Handling environmental Big Data (volume, velocity, variety); integrating multi-scale data; model complexity and accuracy; translating analysis into policy; computational and resource limitations (Dritsas and Trigka, 2025; Phila Sibandze et al., 2024; Tu et al., 2023).
- **Solutions:** Integration of GIS and Remote Sensing, use of AI and ML for classification and prediction, cloud platforms, data sharing through geoportals, advanced approaches for modelling (Dritsas and Trigka, 2025; Arun and Kumar, 2024; Jiang et al., 2019).
- **Future Directions:** Advanced GeoAI for automated monitoring and prediction, widely deployed drone/sensor networks, ecosystem digital twins, near-real-time warning systems, and better socio-ecological integration (arXiv, 2024b; Dritsas and Trigka, 2025; MGISS, 2024).

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